

Process Activity Task Matrix (PATM)

Process	Activity	Task	Source Document	Frequency	Implementer	reviewer
Project Management (PM)	Initiation	Confirm scope (triage + CRM + DB + history reasoning)	Statement of Work		Ahmed Hassan	Amr Samir
		Define high-level requirements & MVP baseline	Requirements List		Mohmed Maher	Aly Essam
		Establish repo baseline + branching + tagging	CM Baseline v0.1		Ahmed Maher	Adham Amir
Planning	Project Planning	Define milestones + DoD	Project Plan		Aly Essam	Ahmed Hassan
		Risk analysis (LLM failure, DB privacy, latency)	Risk Plan		Amr Samir	Ahmed Maher
		Define architecture + DB + services	Architecture Spec		Ahmed Maher	Mohmed Maher
		Define test strategy	QA Plan		Ahmed Hassan	Adham Amir
		Define review checklist	Review Plan		Mohmed Maher	Ahmed Hassan
		Define versioning strategy	CM Plan		Adham Amir	Aly Essam
Execution	Development	Implement Reasonerfactory	Code + Logs		Amr Samir	Mohmed Maher
		Implement RAG evaluation	rag_eval_report.json		Adham Amir	Amr Samir

		Implement DB schema + migrations	DB Schema		Ahmed Hassan	Amr Samir
		Implement CRM endpoints	API Endpoints		Mohmed Maher	Aly Essam
		Implement patient-history provider	Module		Amr Samir	Ahmed Maher
Monitoring						
Closure						

Entity Type Specification (ETS)

Entity: Patient

Attribute	Type	Size	Validations
*PatientID	Int	4	PK, Not Null
Name	Varchar	50	Not Null
Age	Int	2	Positive
Phone	Varchar	15	Not Null
Address	Varchar	100	Not Null

Entity: Doctor

Attribute	Type	Size	Validations
*DoctorID	Int	4	PK, Not Null
Name	Varchar	50	Not Null
Specialization	Varchar	50	Not Null
Phone	Varchar	15	Not Null

Entity: Appointment

Attribute	Type	Size	Validations
*AppointmentID	Int	4	PK, Not Null
PatientID	Int	4	FK -> Patient
DoctorID	Int	4	FK -> Doctor
Date	Date	-	Not Null
Status	Varchar	20	Scheduled/Completed

Entity: Visit

Attribute	Type	Size	Validations
*VisitID	Int	4	PK, Not Null
PatientID	Int	4	FK -> Patient
DoctorID	Int	4	FK -> Doctor
Diagnosis	Varchar	100	-
Notes	Varchar	200	-

Relational Design:

Patient (PatientID, Name, Age, Phone, Address)

Doctor (DoctorID, Name, Specialization, Phone)

Appointment (AppointmentID, PatientID (FK), DoctorID (FK), Date, Status)

Visit (VisitID, PatientID (FK), DoctorID (FK), Diagnosis, Notes)

Integrity Rules:

Associative Integrity Rules

Owner	Owner	Owned	Rule
Patient	Doctor	Appointment	Insertion: Insertion of a tuple in Appointment implies existence of PatientID and DoctorID in Patient and Doctor respectively Deletion: Deletion of a tuple from Patient implies deletion of all tuples having PatientID = Appointment.PatientID Deletion: Deletion of a tuple from Doctor implies deletion of all tuples having DoctorID = Appointment.DoctorID

Owner	Owner	Owned	Rule
Patient	Doctor	Visit	Insertion: Insertion of a tuple in Visit implies existence of PatientID and DoctorID in Patient and Doctor respectively Deletion: Deletion of a tuple from Patient implies deletion of all tuples having PatientID = Visit.PatientID Deletion: Deletion of a tuple from Doctor implies deletion of all tuples having DoctorID = Visit.DoctorID

Referential Integrity Rule

- **Appointment.PatientID → Patient.PatientID**
- **Appointment.DoctorID → Doctor.DoctorID**
- **Visit.PatientID → Patient.PatientID**
- **Visit.DoctorID → Doctor.DoctorID**

Meaning:

- You **cannot insert** Appointment or Visit unless Patient & Doctor exist
- You **cannot reference non-existing IDs**